worksheet cos.

Date: 4-2-25 Lec # 10 Day: M T W T F S

z function of x, y.

Chain rule : Case # 02 :-

Suppose $z = f(x,y)$ is a differentiable function of x and y, where $x = g(s,t)$ and $y = h(s,t)$ are differentiable function of s and t. then

$$\boxed{\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \\ &\text{and} \\ \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} \end{aligned}}$$

Exp:- $z = f(x,y) = e^x \sin y$,

where $x = st^2$, $y = s^2t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$?

Sol:-

(i) $\frac{\partial z}{\partial s} = ?$

$\frac{\partial z}{\partial x} = e^x \sin y$, $\frac{\partial x}{\partial s} = t^2$,

$\frac{\partial z}{\partial y} = e^x \cos y$, $\frac{\partial y}{\partial s} = 2st$

$$\boxed{\frac{\partial z}{\partial s} = (e^x \sin y)(t^2) + (e^x \cos y)(2st)}$$

(ii) $\frac{\partial z}{\partial t} = ?$

$\frac{\partial z}{\partial x} = e^{x} \sin y$, $\frac{\partial x}{\partial t} = 2st$

$\frac{\partial z}{\partial y} = e^{x} \cos y$, $\frac{\partial y}{\partial t} = s^{2}$

$$\boxed{\frac{\partial z}{\partial t} = (e^{x} \sin y)(2st) + (e^{x} \cos y)(s^{2})}$$

Exp:- find $\frac{dy}{dx}$, if $x^{3} + y^{3} = 6xy$.

$$\boxed{\frac{dy}{dx} = -\frac{F_x}{F_y}}$$

$$\frac{dy}{dx} = -\frac{3x^{2} - 6y}{3y^{2} - 6x}$$

$$\frac{dy}{dx} = \frac{\cancel{3}(x^{2} - 2y)}{\cancel{3}(y^{2} - 2x)}$$

$$\boxed{\frac{dy}{dx} = \frac{x^{2} - 2y}{y^{2} - 2x}}$$